THE COSIMPLICIAL DOLD-KAN CONSTRUCTION CARRIES HOMOTOPIC MAPS TO HOMOTOPIC MAPS

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-homotopy-s-Q (source lines 5278-5288); the proof environment begins on the second line after it and is its own proof, closing at source line 5303.

Editorial note. The statement and the proof invoke two author results of the same chapter by label; both are retained as context spans and the generator expands the context transitively (budget 800 lines for this card), while reference targets in other chapters that carry canonical project tags are auto-declared as non-load-bearing source pointers.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Methods chapter, counted once: the cosimplicial Dold-Kan construction carries homotopic maps to homotopic maps on s and on Q.

Allowed standard prerequisites (editorial).

- Cosimplicial objects and their cochain complexes. For a cosimplicial object U of an additive category, the associated cochain complex s(U) and, over an abelian category, the complex Q(U).. They are the objects of the statement (source lines 5283-5287).
- Homotopy and the opposite category. Homotopic morphisms of cosimplicial objects correspond to homotopic morphisms in the opposite category via the contravariant functor A to A (the author's lemma-push-outs-simplicial-object-w-section, retained as context).. It transports the hypothesis to the opposite category (source lines 5291-5294).
- The simplicial statement and the identification of the opposites. Homotopic maps of simplicial objects induce homotopic maps on s and on N (the author's lemma-homotopy-s-N, retained as context), together with the identities s(a') = (s(a))' and Q(U)' = N(U').. It supplies the homotopy in the opposite category and the two identifications (source lines 5295-5302).

Source reference. Original lemma 010V: sources/stacks-chow/tags.txt, source lines 1341–1341. Cross-chapter pointer whose target carries the canonical tag 010V in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Date: stacks-project:simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 010W: sources/stacks-chow/tags.txt, source lines 1342–1342. Cross-chapter pointer whose target carries the canonical tag 010W in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of a category $\mathcal{C}$ with pushouts. Assume there is a morphism $s : Y \to X$ with $s \circ f = id_X$. Consider the cosimplicial object U constructed in Example 0.3 starting with f. The morphism $U \to U$ which in each degree is the self map of $Y \amalg_X \ldots \amalg_X Y$ given by $f \circ s$ on each factor is homotopic to the identity on U. In particular, U is homotopy equivalent to the constant cosimplicial object X.*

Proof. This lemma is dual to Lemma 0.5. Hence this lemma follows on applying Lemma 0.4. □

Lemma 0.2. *Let $\mathcal{A}$ be an additive category. Let $a, b : U \to V$ be morphisms of simplicial objects of $\mathcal{A}$. If a, b are homotopic, then $s(a), s(b) : s(U) \to s(V)$ are homotopic maps of chain complexes. If $\mathcal{A}$ is abelian, then also $N(a), N(b) : N(U) \to N(V)$ are homotopic maps of chain complexes.*

Proof. We may choose a sequence $a = a_0, a_1, \ldots, a_n = b$ of morphisms from U to V such that for each $i = 1, \ldots, n$ either there is a homotopy from a_i to a_{i-1} or there is a homotopy from a_{i-1} to a_i. The calculation above shows that in this case either $s(a_i)$ is homotopic to $s(a_{i-1})$ as a map of chain complexes or $s(a_{i-1})$ is homotopic to $s(a_i)$ as a map of chain complexes. Of course, these things are equivalent and moreover being homotopic is an equivalence relation on the set of maps of chain complexes, see Homology, Section 010V. This proves that $s(a)$ and $s(b)$ are homotopic as maps of chain complexes.

Next, we turn to $N(a)$ and $N(b)$. It follows from Lemma 0.6 that $N(a)$, $N(b)$ are compositions

$$N(U) \to s(U) \to s(V) \to N(V)$$

where we use $s(a)$, $s(b)$ in the middle. Hence the assertion follows from Homology, Lemma 010W. □

Example 0.3. Suppose that $X \to Y$ is a morphism of C such that all the pushouts $Y \amalg_X Y \amalg_X \ldots \amalg_X Y$ exist. Then we set U_n equal to the $(n+1)$-fold pushout, and we let $\varphi : [n] \to [m]$ correspond to the map

$$(y \text{ in } i\text{th component}) \mapsto (y \text{ in } \varphi(i)\text{th component})$$

on "coordinates". In other words, the map $U_1 = Y \amalg_X Y \to U_0 = Y$ is the identity on each component. The two maps $U_0 = Y \to U_1 = Y \amalg_X Y$ are the two coprojections.

Lemma 0.4. *Let $\mathcal{C}$ be a category. Suppose that U and V are two cosimplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be morphisms of cosimplicial objects. Recall that U, V correspond to simplicial objects U', V' of $\mathcal{C}^{opp}$. Moreover a, b correspond to morphisms $a', b' : V' \to U'$. The following are equivalent*

1. *There exists a homotopy $h = \{h_{n,\alpha}\}$ from a to b as in Remark 0.11.*
2. *There exists a homotopy $h = \{h_{n,i}\}$ from a' to b' as in Remark 0.10.*

Thus a is homotopic to b as in Remark 0.11 if and only if a' is homotopic to b' as in Remark 0.10.

Proof. In case $\mathcal{C}$ has finite products, then $\mathcal{C}^{opp}$ has finite coproducts and we may use Definitions 0.8 and 0.7 instead of Remarks 0.11 and 0.10. In this case $h : U \to \mathrm{Hom}(\Delta[1], V)$ is the same as a morphism $h' : \mathrm{Hom}(\Delta[1], V)' \to U'$. Since products and coproducts get switched too, it is immediate that $(\mathrm{Hom}(\Delta[1], V))' = V' \times \Delta[1]$. Moreover, the "primed" version of the morphisms $e_0, e_1 : \mathrm{Hom}(\Delta[1], V) \to V$ are the morphisms $e_0, e_1 : V' \to \Delta[1] \times V$. Thus $e_0 \circ h = a$ translates into $h' \circ e_0 = a'$ and similarly $e_1 \circ h = b$ translates into $h' \circ e_1 = b'$. This proves the lemma in this case.

In the general case, one needs to translate the relations given by (0.8.1) into the relations given in Lemma 0.9. We omit the details.

The final assertion is formal from the equivalence of (1) and (2). □

Lemma 0.5. *Let $f : Y \to X$ be a morphism of a category $\mathcal{C}$ with fibre products. Assume f has a section s. Consider the simplicial object U constructed in Example 0.12 starting with f. The morphism $U \to U$ which in each degree is the self map $(s \circ f)^{n+1}$ of $Y \times_X \ldots \times_X Y$ given by $s \circ f$ on each factor is homotopic to the identity on U. In particular, U is homotopy equivalent to the constant simplicial object X.*

Proof. Set $g^0 = \mathrm{id}_Y$ and $g^1 = s \circ f$. We use the morphisms

$$\begin{aligned} Y \times_X \ldots \times_X Y \times \mathrm{Mor}([n], [1]) &\to Y \times_X \ldots \times_X Y \\ (y_0, \ldots, y_n) \times \alpha &\mapsto (g^{\alpha(0)}(y_0), \ldots, g^{\alpha(n)}(y_n)) \end{aligned}$$

where we use the functor of points point of view to define the maps. Another way to say this is to say that $h_{n,0} = \mathrm{id}$, $h_{n,n+1} = (s \circ f)^{n+1}$ and $h_{n,i} = \mathrm{id}_Y^{i+1} \times (s \circ f)^{n+1-i}$. We leave it to the reader to show that these satisfy the relations of Lemma 0.9. Hence they define the desired homotopy. See also Remark 0.10 which shows that we do not need to assume anything else on the category $\mathcal{C}$. □

Lemma 0.6. *Let $\mathcal{A}$ be an abelian category. Let U be a simplicial object of $\mathcal{A}$. The canonical morphism of chain complexes $N(U) \to s(U)$ is split. In fact,*

$$s(U) = N(U) \oplus D(U)$$

for some complex $D(U)$. The construction $U \mapsto D(U)$ is functorial.

Proof. Define $D(U)_n$ to be the image of

$$\bigoplus\nolimits_{\varphi:[n]\to[m] \text{ surjective, } m<n} N(U_m) \xrightarrow{\bigoplus U(\varphi)} U_n$$

which is a subobject of U_n complementary to $N(U_n)$ according to Lemma 0.14 and Definition 0.13. We show that $D(U)$ is a subcomplex. Pick a surjective map $\varphi : [n] \to [m]$ with $m < n$ and consider the composition

$$N(U_m) \xrightarrow{U(\varphi)} U_n \xrightarrow{d_n} U_{n-1}.$$

This composition is the sum of the maps

$$N(U_m) \xrightarrow{U(\varphi \circ \delta_i^n)} U_{n-1}$$

with sign $(-1)^i$, $i = 0, \ldots, n$.

First we will prove by ascending induction on m, $0 \leq m < n-1$ that all the maps $U(\varphi \circ \delta_i^n)$ map $N(U_m)$ into $D(U)_{n-1}$. (The case $m = n-1$ is treated below.) Whenever the map $\varphi \circ \delta_i^n : [n-1] \to [m]$ is surjective then the image of $N(U_m)$

under $U(\varphi \circ \delta_i^n)$ is contained in $D(U)_{n-1}$ by definition. If $\varphi \circ \delta_i^n : [n-1] \to [m]$ is not surjective, set $j = \varphi(i)$ and observe that i is the unique index whose image under φ is j. We may write $\varphi \circ \delta_i^n = \delta_j^m \circ \psi \circ \delta_i^n$ for some $\psi : [n-1] \to [m-1]$. Hence $U(\varphi \circ \delta_i^n) = U(\psi \circ \delta_i^n) \circ d_j^m$ which is zero on $N(U_m)$ unless $j = m$. If $j = m$, then $d_m^m(N(U_m)) \subset N(U_{m-1})$ and hence $U(\varphi \circ \delta_i^n)(N(U_m)) \subset U(\psi \circ \delta_i^n)(N(U_{m-1}))$ and we win by induction hypothesis.

To finish proving that $D(U)$ is a subcomplex we still have to deal with the composition

$$N(U_m) \xrightarrow{U(\varphi)} U_n \xrightarrow{d_n} U_{n-1}.$$

in case $m = n-1$. In this case $\varphi = \sigma_j^{n-1}$ for some $0 \leq j \leq n-1$ and $U(\varphi) = s_j^{n-1}$. Thus the composition is given by the sum

$$\sum (-1)^i d_i^n \circ s_j^{n-1}$$

Recall from Remark 0.15 that $d_j^n \circ s_j^{n-1} = d_{j+1}^n \circ s_j^{n-1} = \mathrm{id}$ and these drop out because the corresponding terms have opposite signs. The map $d_n^n \circ s_j^{n-1}$, if $j < n-1$, is equal to $s_j^{n-2} \circ d_{n-1}^{n-1}$. Since d_{n-1}^{n-1} maps $N(U_{n-1})$ into $N(U_{n-2})$, we see that the image $d_n^n(s_j^{n-1}(N(U_{n-1}))$ is contained in $s_j^{n-2}(N(U_{n-2}))$ which is contained in $D(U_{n-1})$ by definition. For all other combinations of (i, j) we have either $d_i^n \circ s_j^{n-1} = s_{j-1}^{n-2} \circ d_i^{n-1}$ (if $i < j$), or $d_i^n \circ s_j^{n-1} = s_j^{n-2} \circ d_{i-1}^{n-1}$ (if $n > i > j+1$) and in these cases the map is zero because of the definition of $N(U_{n-1})$. □

Definition 0.7. Let $\mathcal{C}$ be a category having finite coproducts. Suppose that U and V are two simplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be two morphisms.

(1) We say a morphism
$$h : U \times \Delta[1] \longrightarrow V$$
is a *homotopy from a to b* if $a = h \circ e_0$ and $b = h \circ e_1$.
(2) We say the morphisms a and b are *homotopic* or are *in the same homotopy class* if there exists a sequence of morphisms $a = a_0, a_1, \ldots, a_n = b$ from U to V such that for each $i = 1, \ldots, n$ there either exists a homotopy from a_{i-1} to a_i or there exists a homotopy from a_i to a_{i-1}.

Definition 0.8. Let $\mathcal{C}$ be a category having finite products. Let U and V be two cosimplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be two morphisms of cosimplicial objects of $\mathcal{C}$.

(1) We say a morphism
$$h : U \longrightarrow \mathrm{Hom}(\Delta[1], V)$$
such that $a = e_0 \circ h$ and $b = e_1 \circ h$ is a *homotopy from a to b*.
(2) We say a and b are *homotopic* or are *in the same homotopy class* if there exists a sequence $a = a_0, a_1, \ldots, a_n = b$ of morphisms from U to V such that for each $i = 1, \ldots, n$ there either exists a homotopy from a_i to a_{i-1} or there exists a homotopy from a_{i-1} to a_i.

$$h_{m,\alpha} \circ U(f) = V(f) \circ h_{n,\alpha \circ f} \tag{0.8.1}$$

Lemma 0.9. *In the situation above, we have the following relations:*

(1) *We have* $h_{n,0} = b_n$ *and* $h_{n,n+1} = a_n$.
(2) *We have* $d^n_j \circ h_{n,i} = h_{n-1,i-1} \circ d^n_j$ *for* $i > j$.
(3) *We have* $d^n_j \circ h_{n,i} = h_{n-1,i} \circ d^n_j$ *for* $i \leq j$.
(4) *We have* $s^n_j \circ h_{n,i} = h_{n+1,i+1} \circ s^n_j$ *for* $i > j$.
(5) *We have* $s^n_j \circ h_{n,i} = h_{n+1,i} \circ s^n_j$ *for* $i \leq j$.

Conversely, given a system of maps $h_{n,i}$ *satisfying the properties listed above, then these define a morphism* h *which is a homotopy from* a *to* b.

Proof. Omitted. You can prove the last statement using the fact, see Lemma 0.16 that to give a morphism of simplicial objects is the same as giving a sequence of morphisms h_n commuting with all d^n_j and s^n_j. □

Remark 0.10. Let $\mathcal{C}$ be any category (no assumptions whatsoever). Let U and V be simplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be morphisms of simplicial objects of $\mathcal{C}$. A *homotopy from* a *to* b is given by morphisms[1] $h_{n,i} : U_n \to V_n$, for $n \geq 0$, $i = 0, \ldots, n+1$ satisfying the relations of Lemma 0.9. As in Definition 0.7 we say the morphisms a and b are *homotopic* if there exists a sequence of morphisms $a = a_0, a_1, \ldots, a_n = b$ from U to V such that for each $i = 1, \ldots, n$ there either exists a homotopy from a_{i-1} to a_i or there exists a homotopy from a_i to a_{i-1}. Clearly, if $F : \mathcal{C} \to \mathcal{C}'$ is any functor and $\{h_{n,i}\}$ is a homotopy from a to b, then $\{F(h_{n,i})\}$ is a homotopy from $F(a)$ to $F(b)$. Similarly, if a and b are homotopic, then $F(a)$ and $F(b)$ are homotopic. Since the lemma says that the newer notion is the same as the old one in case finite coproduct exist, we deduce in particular that functors preserve the original notion whenever both categories have finite coproducts.

Remark 0.11. Let $\mathcal{C}$ be any category (no assumptions whatsoever). Let U and V be cosimplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be morphisms of cosimplicial objects of $\mathcal{C}$. A *homotopy from* a *to* b is given by morphisms $h_{n,\alpha} : U_n \to V_n$, for $n \geq 0$, $\alpha \in \Delta[1]_n$ satisfying (0.8.1) for all morphisms f of Δ and such that $a_n = h_{n,0:[n]\to[1]}$ and $b_n = h_{n,1:[n]\to[1]}$ for all $n \geq 0$. As in Definition 0.8 we say the morphisms a and b are *homotopic* if there exists a sequence of morphisms $a = a_0, a_1, \ldots, a_n = b$ from U to V such that for each $i = 1, \ldots, n$ there either exists a homotopy from a_{i-1} to a_i or there exists a homotopy from a_i to a_{i-1}. Clearly, if $F : \mathcal{C} \to \mathcal{C}'$ is any functor and $\{h_{n,i}\}$ is a homotopy from a to b, then $\{F(h_{n,i})\}$ is a homotopy from $F(a)$ to $F(b)$. Similarly, if a and b are homotopic, then $F(a)$ and $F(b)$ are homotopic. This new notion is the same as the old one in case finite products exist. We deduce in particular that functors preserve the original notion whenever both categories have finite products.

Example 0.12. Suppose that $Y \to X$ is a morphism of $\mathcal{C}$ such that all the fibred products $Y \times_X Y \times_X \ldots \times_X Y$ exist. Then we set U_n equal to the $(n+1)$-fold fibre product, and we let $\varphi : [n] \to [m]$ correspond to the map (on "coordinates") $(y_0, \ldots, y_m) \mapsto (y_{\varphi(0)}, \ldots, y_{\varphi(n)})$. In other words, the map $U_0 = Y \to U_1 = Y \times_X Y$ is the diagonal map. The two maps $U_1 = Y \times_X Y \to U_0 = Y$ are the projection maps.

Definition 0.13. Let $\mathcal{C}$ be a category which admits finite nonempty coproducts. We say a simplicial object U of $\mathcal{C}$ is *split* if there exist subobjects $N(U_m)$ of U_m,

[1] In the literature, often the maps $h_{n+1,i} \circ s_i : U_n \to V_{n+1}$ are used instead of the maps $h_{n,i}$. Of course the relations these maps satisfy are different from the ones in Lemma 0.9.

$m \geq 0$ with the property that

$$\coprod_{\varphi:[n]\to[m] \text{ surjective}} N(U_m) \longrightarrow U_n \tag{0.13.1}$$

is an isomorphism for all $n \geq 0$. If U is an r-truncated simplicial object of $\mathcal{C}$ then we say U is *split* if there exist subobjects $N(U_m)$ of U_m, $r \geq m \geq 0$ with the property that (0.13.1) is an isomorphism for $r \geq n \geq 0$.

Lemma 0.14. *Let $\mathcal{A}$ be an abelian category. Let U be a simplicial object in $\mathcal{A}$. Then U has a splitting obtained by taking $N(U_0) = U_0$ and for $m \geq 1$ taking*

$$N(U_m) = \bigcap\nolimits_{i=0}^{m-1} \mathrm{Ker}(d_i^m).$$

Moreover, this splitting is functorial on the category of simplicial objects of $\mathcal{A}$.

Proof. For any object A of $\mathcal{A}$ we obtain a simplicial abelian group $\mathrm{Mor}_{\mathcal{A}}(A, U)$. Each of these are canonically split by Lemma 0.17. Moreover,

$$N(\mathrm{Mor}_{\mathcal{A}}(A, U_m)) = \bigcap\nolimits_{i=0}^{m-1} \mathrm{Ker}(d_i^m) = \mathrm{Mor}_{\mathcal{A}}(A, N(U_m)).$$

Hence we see that the morphism (0.13.1) becomes an isomorphism after applying the functor $\mathrm{Mor}_{\mathcal{A}}(A, -)$ for any object of $\mathcal{A}$. Hence it is an isomorphism by the Yoneda lemma. □

Remark 0.15. By abuse of notation we sometimes write $d_i : U_n \to U_{n-1}$ instead of d_i^n, and similarly for $s_i : U_n \to U_{n+1}$. The relations among the morphisms d_i^n and s_i^n may be expressed as follows:

(1) If $i < j$, then $d_i \circ d_j = d_{j-1} \circ d_i$.
(2) If $i < j$, then $d_i \circ s_j = s_{j-1} \circ d_i$.
(3) We have $\mathrm{id} = d_j \circ s_j = d_{j+1} \circ s_j$.
(4) If $i > j + 1$, then $d_i \circ s_j = s_j \circ d_{i-1}$.
(5) If $i \leq j$, then $s_i \circ s_j = s_{j+1} \circ s_i$.

This means that whenever the compositions on both the left and the right are defined then the corresponding equality should hold.

Lemma 0.16. *The category Δ is the universal category with objects $[n]$, $n \geq 0$ and morphisms δ_j^n and σ_j^n such that (a) every morphism is a composition of these morphisms, (b) the relations listed in Lemma 0.18 are satisfied, and (c) any relation among the morphisms is a consequence of those relations.*

Proof. Omitted. □

Lemma 0.17. *Let U be a simplicial abelian group. Then U has a splitting obtained by taking $N(U_0) = U_0$ and for $m \geq 1$ taking*

$$N(U_m) = \bigcap\nolimits_{i=0}^{m-1} \mathrm{Ker}(d_i^m).$$

Moreover, this splitting is functorial on the category of simplicial abelian groups.

Proof. By induction on n we will show that the choice of $N(U_m)$ in the lemma guarantees that (0.13.1) is an isomorphism for $m \leq n$. This is clear for $n = 0$. In the rest of this proof we are going to drop the superscripts from the maps d_i and s_i in order to improve readability. We will also repeatedly use the relations from Remark 0.15.

First we make a general remark. For $0 \leq i \leq m$ and $z \in U_m$ we have $d_i(s_i(z)) = z$. Hence we can write any $x \in U_{m+1}$ uniquely as $x = x' + x''$ with $d_i(x') = 0$ and $x'' \in \mathrm{Im}(s_i)$ by taking $x' = (x - s_i(d_i(x)))$ and $x'' = s_i(d_i(x))$. Moreover, the element $z \in U_m$ such that $x'' = s_i(z)$ is unique because s_i is injective.

Here is a procedure for decomposing any $x \in U_{n+1}$. First, write $x = x_0 + s_0(z_0)$ with $d_0(x_0) = 0$. Next, write $x_0 = x_1 + s_1(z_1)$ with $d_1(x_1) = 0$. Continue like this to get

$$\begin{aligned} x &= x_0 + s_0(z_0), \\ x_0 &= x_1 + s_1(z_1), \\ x_1 &= x_2 + s_2(z_2), \\ \ldots &\quad \ldots \quad \ldots \\ x_{n-1} &= x_n + s_n(z_n) \end{aligned}$$

where $d_i(x_i) = 0$ for all $i = n, \ldots, 0$. By our general remark above all of the x_i and z_i are determined uniquely by x. We claim that $x_i \in \mathrm{Ker}(d_0) \cap \mathrm{Ker}(d_1) \cap \ldots \cap \mathrm{Ker}(d_i)$ and $z_i \in \mathrm{Ker}(d_0) \cap \ldots \cap \mathrm{Ker}(d_{i-1})$ for $i = n, \ldots, 0$. Here and in the following an empty intersection of kernels indicates the whole space; i.e., the notation $z_0 \in \mathrm{Ker}(d_0) \cap \ldots \cap \mathrm{Ker}(d_{i-1})$ when $i = 0$ means $z_0 \in U_n$ with no restriction.

We prove this by ascending induction on i. It is clear for $i = 0$ by construction of x_0 and z_0. Let us prove it for $0 < i \leq n$ assuming the result for $i-1$. First of all we have $d_i(x_i) = 0$ by construction. So pick a j with $0 \leq j < i$. We have $d_j(x_{i-1}) = 0$ by induction. Hence

$$0 = d_j(x_{i-1}) = d_j(x_i) + d_j(s_i(z_i)) = d_j(x_i) + s_{i-1}(d_j(z_i)).$$

The last equality by the relations of Remark 0.15. These relations also imply that $d_{i-1}(d_j(x_i)) = d_j(d_i(x_i)) = 0$ because $d_i(x_i) = 0$ by construction. Then the uniqueness in the general remark above shows the equality $0 = x' + x'' = d_j(x_i) + s_{i-1}(d_j(z_i))$ can only hold if both terms are zero. We conclude that $d_j(x_i) = 0$ and by injectivity of s_{i-1} we also conclude that $d_j(z_i) = 0$. This proves the claim.

The claim implies we can uniquely write

$$x = s_0(z_0) + s_1(z_1) + \ldots + s_n(z_n) + x_n$$

with $x_n \in N(U_{n+1})$ and $z_i \in \mathrm{Ker}(d_0) \cap \ldots \cap \mathrm{Ker}(d_{i-1})$. We can reformulate this as saying that we have found a direct sum decomposition

$$U_{n+1} = N(U_{n+1}) \oplus \bigoplus\nolimits_{i=0}^{i=n} s_i\Big(\mathrm{Ker}(d_0) \cap \ldots \cap \mathrm{Ker}(d_{i-1})\Big)$$

with the property that

$$\mathrm{Ker}(d_0) \cap \ldots \cap \mathrm{Ker}(d_j) = N(U_{n+1}) \oplus \bigoplus\nolimits_{i=j+1}^{i=n} s_i\Big(\mathrm{Ker}(d_0) \cap \ldots \cap \mathrm{Ker}(d_{i-1})\Big)$$

for $j = 0, \ldots, n$. The result follows from this statement as follows. Each of the z_i in the expression for x can be written uniquely as

$$z_i = s_i(z'_{i,i}) + \ldots + s_{n-1}(z'_{i,n-1}) + z_{i,0}$$

with $z_{i,0} \in N(U_n)$ and $z'_{i,j} \in \mathrm{Ker}(d_0) \cap \ldots \cap \mathrm{Ker}(d_{j-1})$. The first few steps in the decomposition of z_i are zero because z_i already is in the kernel of $d_0, \ldots, d_i$. This

in turn uniquely gives

$$x = x_n + s_0(z_{0,0}) + s_1(z_{1,0}) + \ldots + s_n(z_{n,0}) + \sum\nolimits_{0 \leq i \leq j \leq n-1} s_i(s_j(z'_{i,j})).$$

Continuing in this fashion we see that we in the end obtain a decomposition of x as a sum of terms of the form

$$s_{i_1} s_{i_2} \ldots s_{i_k}(z)$$

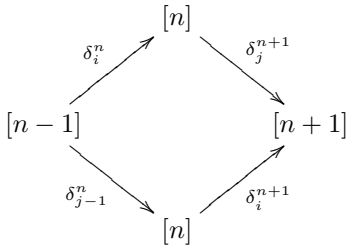
with $0 \leq i_1 \leq i_2 \leq \ldots \leq i_k \leq n-k+1$ and $z \in N(U_{n+1-k})$. This is exactly the required decomposition, because any surjective map $[n+1] \to [n+1-k]$ can be uniquely expressed in the form

$$\sigma^{n-k}_{i_k} \ldots \sigma^{n-1}_{i_2} \sigma^n_{i_1}$$

with $0 \leq i_1 \leq i_2 \leq \ldots \leq i_k \leq n-k+1$. □

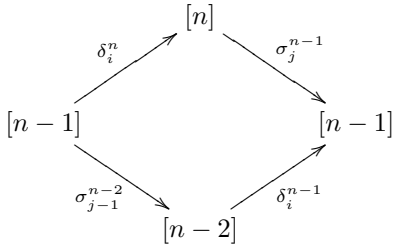
Lemma 0.18. *The morphisms δ^n_j and σ^n_j satisfy the following relations.*

(1) *If $0 \leq i < j \leq n+1$, then $\delta^{n+1}_j \circ \delta^n_i = \delta^{n+1}_i \circ \delta^n_{j-1}$. In other words the diagram*



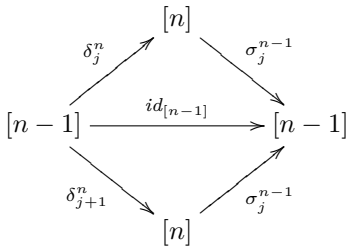
commutes.

(2) *If $0 \leq i < j \leq n-1$, then $\sigma^{n-1}_j \circ \delta^n_i = \delta^{n-1}_i \circ \sigma^{n-2}_{j-1}$. In other words the diagram*



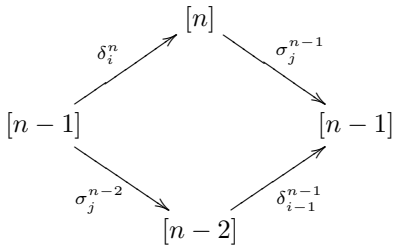
commutes.

(3) *If $0 \leq j \leq n-1$, then $\sigma^{n-1}_j \circ \delta^n_j = id_{[n-1]}$ and $\sigma^{n-1}_j \circ \delta^n_{j+1} = id_{[n-1]}$. In other words the diagram*



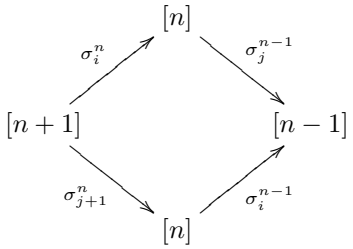
commutes.

(4) *If* $0 < j+1 < i \leq n$, *then* $\sigma_j^{n-1} \circ \delta_i^n = \delta_{i-1}^{n-1} \circ \sigma_j^{n-2}$. *In other words the diagram*



commutes.

(5) *If* $0 \leq i \leq j \leq n-1$, *then* $\sigma_j^{n-1} \circ \sigma_i^n = \sigma_i^{n-1} \circ \sigma_{j+1}^n$. *In other words the diagram*



commutes.

Proof. Omitted. □

Lemma 0.19. *Let* $\mathcal{A}$ *be an additive category. Let* $a, b : U \to V$ *be morphisms of cosimplicial objects of* $\mathcal{A}$. *If* a, b *are homotopic, then* $s(a), s(b) : s(U) \to s(V)$ *are homotopic maps of cochain complexes. If in addition* $\mathcal{A}$ *is abelian, then* $Q(a), Q(b) : Q(U) \to Q(V)$ *are homotopic maps of cochain complexes.*

Proof. Let $(-)' : \mathcal{A} \to \mathcal{A}^{opp}$ be the contravariant functor $A \mapsto A$. By Lemma 0.1 the maps a' and b' are homotopic. By Lemma 0.2 we see that $s(a')$ and $s(b')$ are homotopic maps of chain complexes. Since $s(a') = (s(a))'$ and $s(b') = (s(b))'$ we conclude that also $s(a)$ and $s(b)$ are homotopic by applying the additive contravariant functor $(-)'' : \mathcal{A}^{opp} \to \mathcal{A}$. The result for the Q-complexes follows in the same manner using that $Q(U)' = N(U')$. □